GRAPHIC ERA (DEEMED TO BE UNIVERSITY)

Department of Computer Science and Engineering

TCS-392: INTRODUCTION TO CRYPTOGRAPHY

UNIT–II ASSIGNMENT

Modern Cryptography and Symmetric Encryption

Name: _____ **University Roll No.:** _____

Section: _____ **Date of Submission:** _____

Instructions

- Attempt all questions.
- Show all steps in numerical problems.
- Draw neat diagrams wherever required.
- Answers should include justification and not only final results.
- Use appropriate cryptographic notation wherever applicable.

Assignment Questions

Q1. Explain the concept of **modern cryptography**. Differentiate between classical cryptography and modern cryptography in terms of design principles, key usage, mathematical foundations, and applications.

Q2. Explain the **symmetric encryption model** with a suitable diagram. Clearly describe:

(a) Plaintext

(b) Encryption algorithm

(c) Secret key

(d) Ciphertext

(e) Decryption algorithm

Q3. What is **message confidentiality**? Explain how symmetric encryption can be used to achieve confidentiality. Also discuss the importance of secure key distribution.

Q4. Explain the basic principles of symmetric encryption. Discuss the role of:

(a) Substitution

(b) Permutation

(c) Confusion

(d) Diffusion

(e) Product cipher

Q5. Differentiate between **cryptanalysis** and **brute-force attack**. Explain how key size affects the resistance of a symmetric cipher to brute-force attacks.

Q6. A symmetric cipher uses a key of n bits. Derive:

$$\text{Total Key Space} = 2^n$$

and calculate the total number of possible keys for:

$$n = 8, \quad 16, \quad 32, \quad 56, \quad 128.$$

Also calculate the average number of trials required in a brute-force attack.

Q7. Explain the **Feistel cipher structure** with a neat diagram. Clearly describe the division of the plaintext block into left and right halves.

Q8. Explain the Feistel round equations:

$$L_i = R_{i-1}$$

and

$$R_i = L_{i-1} \oplus F(R_{i-1}, K_i).$$

Explain the role of the round function F and the round key K_i.

Q9. Feistel Numerical: Consider:

$$L_{i-1} = 10110110$$

$$R_{i-1} = 01101001$$

and

$$F(R_{i-1}, K_i) = 11001010.$$

Calculate:

(a) L_i

(b) R_i

(c) The complete output block $L_i R_i$

Q10. Explain why the same Feistel structure can be used for both encryption and decryption. What change is required in the round-key order during decryption?

Q11. Define a **block cipher**. Differentiate between block ciphers and stream ciphers on the basis of:

(a) Data processing

(b) Encryption unit

(c) Key-stream usage

(d) Error propagation

(e) Examples

Q12. Explain the general design principles of a symmetric block cipher. Discuss the significance of:

(a) Block size

(b) Key size

(c) Number of rounds

(d) Round function

(e) Subkey generation

Q13. What is **Simplified DES (S-DES)**? Explain why simplified block ciphers are useful for understanding the design principles of practical symmetric encryption algorithms.

Q14. Explain the major operations performed in a DES-type round function:

(a) Expansion

(b) XOR with round key

(c) S-box substitution

(d) Permutation

Q15. Simplified Feistel Numerical: Given:

$$L_{i-1} = 011100$$

$$R_{i-1} = 100110$$

and

$$f(R_{i-1}, K_i) = 000100,$$

calculate:

(a) L_i

(b) R_i

(c) $L_i R_i$

Q16. Explain the purpose of **S-boxes** in symmetric block ciphers. Why is nonlinear substitution important for cryptographic security?

Q17. Explain **Double DES**. If $E_K(P)$ denotes encryption of plaintext P using key K, explain the encryption operation:

$$C = E_{K_2}(E_{K_1}(P))$$

and write the corresponding decryption operation.

Q18. Explain the **meet-in-the-middle attack** on Double DES. Why does Double DES not provide the full security expected from two independent 56-bit keys?

Q19. Prepare a detailed comparison between **Single DES and Double DES** in terms of:

(a) Number of keys

(b) Number of encryption operations

(c) Nominal key space

(d) Effective security

(e) Vulnerability to attacks

Q20. Define a **stream cipher**. Explain how a stream cipher produces ciphertext using:

$$C_i = P_i \oplus K_i.$$

Also explain why decryption can be performed using:

$$P_i = C_i \oplus K_i.$$

Q21. Stream Cipher Numerical: Given:

$$P = 10110010$$

and

$$K = 01101001,$$

calculate:

$$C = P \oplus K.$$

Show all XOR steps.

Q22. Stream Cipher Decryption Numerical: Given:

$$C = 11011011$$

and

$$K = 01101001,$$

calculate:

$$P = C \oplus K.$$

Verify that the original plaintext is recovered.

Q23. Explain the **RC4 algorithm**. Discuss its historical development, byte-oriented operation, variable key size, state vector, and use as a stream cipher.

Q24. Explain the initialization of RC4. Clearly describe the purpose of the state vector S and temporary vector T using:

$$S[i] = i$$

and

$$T[i] = K[i \bmod \text{keylen}].$$

Q25. Explain the **RC4 Key Scheduling Algorithm (KSA)**. Describe the significance of:

$$j = (j + S[i] + T[i]) \bmod 256$$

and

$$\text{Swap}(S[i], S[j]).$$

Q26. RC4 KSA Numerical: Consider a toy RC4 system with:

$$S = [0, 1, 2, 3, 4, 5, 6, 7]$$

and key:

$$K = [3, 1, 4].$$

Construct the temporary vector T of length 8 using:

$$T[i] = K[i \bmod 3].$$

Then perform the first three KSA iterations and show:

(a) Value of i

(b) Value of j

(c) Elements swapped

(d) Updated state vector S

Q27. Explain the **RC4 Pseudo-Random Generation Algorithm (PRGA)** using:

$$i = (i + 1) \bmod 256$$

$$j = (j + S[i]) \bmod 256$$

$$\text{Swap}(S[i], S[j])$$

$$t = (S[i] + S[j]) \bmod 256$$

$$k = S[t].$$

Explain how the value k is used during encryption.

Q28. Explain why RC4 is no longer considered secure for modern cryptographic applications. Discuss:

(a) Key-stream bias

(b) Weaknesses in early output bytes

(c) Problems in WEP

(d) Historical use in SSL/TLS

(e) Modern deprecation of RC4

Q29. Explain a **True Random Number Generator (TRNG)**. Define entropy and discuss the following entropy sources:

(a) Thermal noise

(b) Sound input

(c) Video or camera input

(d) Disk timing

(e) Keyboard and mouse activity

Also explain why raw physical measurements may require additional processing.

Q30. Differentiate between **True Random Number Generators (TRNGs)** and **Pseudo-random Number Generators (PRNGs)**. Explain:

(a) Deterministic and nondeterministic generation

(b) Role of a seed

(c) Entropy

(d) Skew or bias

(e) Deskewing or conditioning

A random source generates 1000 bits containing 650 ones and 350 zeros. Calculate:

$$P(1)$$

and

$$P(0).$$

Comment on whether the source is balanced or biased, and explain the importance of secure random-number generation in cryptographic keys, nonces, initialization vectors, and digital signatures.

— End of Unit–II Assignment —